

USUS

Product Specifications Document

AI-Powered Simulation Environment for Human Staff Training, Evaluation & Onboarding

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1. Executive Summary

Usus is an AI-powered simulation environment designed to help human staff practice real-world job scenarios and receive structured, objective feedback on their performance. Instead of relying on static training materials, scripts, or occasional role-play sessions, Usus enables continuous, realistic practice in a safe environment where mistakes become learning opportunities.

The core idea is that people learn best by doing. Usus aims to become the training equivalent of a flight simulator — repeatable, measurable, and grounded in reality.

Vision: Usus becomes the standard training simulator for human-facing roles, starting with customer support and expanding into sales, HR, and any profession that requires people skills.

1.1 Product Identity

Attribute	Detail
Product Name	Usus
Product Type	AI Simulation & Training Platform (SaaS + Internal)
Platform	Web Application (Next.js) + Mobile (responsive / future native)
Initial Focus	Customer Support / CRM Personnel
Long-Term Scope	All human-facing roles: Sales, HR, Retail, Healthcare, etc.
Delivery Model	Multi-tenant SaaS with company-specific customization
AI Model Strategy	Model-agnostic (flexible integration with any LLM provider)

1.2 Core Value Proposition

- **Practice by doing:** Staff train through realistic simulations, not slideshows and scripts
- **AI as customer, superior, or peer:** The AI adapts to play any role in the scenario
- **Objective evaluation:** Every interaction scored across multiple dimensions with structured feedback
- **Company-specific customization:** Scenarios tailored to each organization's products, policies, and brand voice
- **Hiring integration:** Evaluate candidates on soft skills before they're hired
- **Continuous improvement:** Structured learning paths that progress from beginner to advanced

2. Problem Statement

Customer support training today is fragmented and ineffective. Teams rely heavily on documentation, onboarding videos, shadowing, and scripts that rarely prepare agents for the complexity of real customer interactions. Evaluation is usually subjective, inconsistent, or delayed, making it difficult for trainees to understand exactly what they did wrong and how to improve.

At the same time, customer expectations are increasing. Users expect empathy, clarity, speed, and accuracy — even in high-stress or emotionally charged situations. Organizations need a way to train support staff that reflects this reality while maintaining consistent standards.

2.1 Current Training Failures

Problem	Impact
Static training materials	No practice under realistic pressure; low retention
Subjective evaluation	Inconsistent standards; favoritism; unclear improvement paths
Delayed feedback	Mistakes repeated before correction; bad habits form
No safe failure space	Staff fear making mistakes; learning is punished, not encouraged
One-size-fits-all scripts	Agents sound robotic; can't handle edge cases or emotional customers
Expensive role-play sessions	Requires senior staff time; doesn't scale; infrequent

3. System Architecture

3.1 Architecture Overview

Usus is built as a modular, multi-tenant system with five core components that work together to deliver the simulation, evaluation, and learning experience.

Component	Function	Key Technology
Simulation Engine	AI plays realistic customers/peers/superiors via text and voice	LLM API (model-agnostic), WebRTC / voice API
Evaluation Engine	Scores performance across 7 dimensions; generates structured feedback	LLM-powered analysis, custom rubric engine
Training Knowledge Base	CS principles, best practices, online reviews, real-world scenarios, company-specific data	Vector database, RAG pipeline
Learning Path Engine	Structured progression from beginner to advanced	State machine, skill graph
Admin Dashboard	Company management, scenario creation, analytics	Next.js, role-based access control

3.2 Tech Stack

Layer	Technology	Rationale
Frontend	Next.js (React)	SSR, fast rendering, strong ecosystem, responsive web + mobile
Backend	Python / FastAPI	High performance async API; strong AI/ML ecosystem
Background Tasks	Celery (or equivalent)	Async job processing for evaluation, report generation
Database	PostgreSQL (primary) + Redis (cache/sessions)	Relational data + fast session management
Vector Storage	Pinecone / pgvector / Weaviate	RAG for knowledge base retrieval
AI / LLM	Model-agnostic (Claude, OpenAI, open-source)	Flexible; swap models without architecture changes
Voice	WebRTC + STT / TTS APIs	Real-time voice simulation (phone call experience)
Hosting	Cloud (AWS / GCP / Vercel)	Scalable, reliable, global distribution
API Standard	OpenAPI / REST	Well-documented, interoperable

3.3 Latency Requirements

Target API response latency: 60ms or less for non-AI endpoints. AI-powered responses (simulation, evaluation) will have higher latency due to LLM processing but should target sub-2-second response times for text and near-real-time for voice streaming.

Interaction Type	Target Latency	Notes
UI interactions / API calls	≤60ms	Standard CRUD, navigation, dashboard loading
Text simulation response	≤2 seconds	LLM generation with streaming enabled
Voice simulation response	Near real-time	Streaming STT → LLM → TTS pipeline
Evaluation generation	≤10 seconds	Post-session analysis; can be async
Report generation	Background	Celery task; user notified on completion

4. Simulation Engine

The simulation engine is the core of Usus. It powers realistic, interactive scenarios where the AI plays the role of a customer, supervisor, or peer — adapting in real-time to trainee responses.

4.1 Simulation Modes

► **REVISED:** *Email and In-Person simulation modes moved to Phase 2. MVP includes Text Chat and Voice Call only.*

Mode	Channel	Description	Phase
Text Chat	Live chat support	Trainee interacts with AI customer via typed messages; simulates live chat or messaging support	MVP
Voice Call	Call center / phone support	Real-time voice conversation; AI speaks as the customer using TTS, trainee speaks naturally via STT	MVP
Email	Email ticketing	Trainee composes email responses to AI-generated customer emails; async evaluation	Phase 2
In-Person	Retail / face-to-face CRM	Text or voice simulation of walk-in scenarios; contextual prompts describe the physical setting	Phase 2

4.2 AI Roles

► **REVISED:** *MVP will feature the Customer role only. Superior/Manager and Peer/Colleague roles are deferred to Phase 2.*

- **Customer (MVP):** The primary role. Simulates real customers with varying personalities, emotions, problems, and difficulty levels.
- **Superior / Manager (Phase 2):** Simulates a supervisor giving instructions, escalating issues, or conducting performance reviews.
- **Peer / Colleague (Phase 2):** Simulates a co-worker for team communication, handoff, or collaboration scenarios.

4.3 Scenario Generation (Hybrid Model)

Usus uses a hybrid approach to scenario generation, combining scripted frameworks with AI improvisation:

Scripted Framework

- Each scenario has a defined structure: context, customer persona, problem type, expected resolution path, and evaluation criteria
- Difficulty level, emotional tone, and compliance requirements are pre-set
- Key checkpoints and decision points are scripted to ensure consistent evaluation

AI Improvisation

- Within the framework, the AI generates natural, varied dialogue — no two runs are identical
- AI adapts to trainee responses: if the trainee de-escalates well, the customer calms down; if poorly, they escalate

- AI introduces realistic curveballs: changing requirements, emotional shifts, tangential questions
- Edge cases and ambiguous problems test critical thinking, not just script adherence

4.4 Customer Personas & Difficulty

► **REVISED:** *The personas below are illustrative starting points. Usus will support a structured process for generating, ranking, and categorizing new personas and scenarios beyond this initial set.*

Persona Type	Behavior	Difficulty
Friendly & Clear	Polite, states problem clearly, patient	Beginner
Confused & Rambling	Unclear problem description, goes off-topic, needs guidance	Intermediate
Frustrated & Impatient	Upset about wait times, past failures, demands immediate resolution	Intermediate
Angry & Hostile	Aggressive language, personal attacks, threatens to escalate	Advanced
Manipulative / Social Engineer	Tries to extract unauthorized info or bypass policy	Advanced
VIP / High-Value	Expects premium treatment, references their importance	Advanced
Emotionally Distressed	Genuine personal hardship connected to the issue	Advanced
Technical Expert	Knows more than the average customer, challenges agent knowledge	Expert

4.5 Scenario Generation, Ranking & Categorization Process

► **NEW:** *New section defining the structured process for expanding beyond the initial persona set.*

To ensure the scenario library remains comprehensive, realistic, and well-organized, Usus follows a structured process for generating, ranking, and categorizing new scenarios:

Generation

- AI-assisted generation: Use LLM analysis of real customer interactions, online reviews, and company-specific data to propose new scenario drafts
- Manual creation: Company admins and platform admins can create custom scenarios via the Scenario Builder
- Community sourcing: Managers can submit scenario ideas that are reviewed and formalized by platform admins

Ranking

- Each scenario is ranked on four criteria: realism (based on real-world frequency), difficulty level, training value (skills exercised), and coverage gap (does it address an underrepresented area?)
- Ranking scores are used to prioritize which scenarios are included in default learning paths
- Scenarios with low realism or low training value are flagged for revision or removal

Categorization

- Every scenario is tagged with: category (e.g., billing, technical support), difficulty level, required skills, persona type, and emotional intensity
- Tags enable filtering, targeted assignment, and gap analysis across the scenario library
- Company admins can add custom tags relevant to their organization

4.6 Scenario Flow

1. Scenario Selection: Trainee picks a scenario from categorized library (or assigned by manager)
2. Context Briefing: System provides background: customer history, product context, any prior interactions
3. Simulation Begins: AI initiates conversation in character; trainee responds via text or voice
4. Interaction Progresses: AI adapts dynamically; introduces complications based on scenario design
5. Resolution or Escalation: Trainee either resolves the issue, escalates, or the scenario times out
6. Scenario Ends: AI exits character; system triggers evaluation
7. Debrief & Feedback: Trainee receives structured performance report (configurable: real-time or post-session)

Important: Users are limited to the scenario they select. To try another scenario, they must explicitly start a new one. This prevents context contamination and ensures clean evaluation.

4.7 Voice Simulation Architecture

Voice simulations replicate a real phone call experience:

1. Trainee speaks: Browser/app captures audio via microphone
2. Speech-to-Text: Audio streamed to STT service for real-time transcription
3. LLM Processing: Transcribed text fed to the simulation engine for AI response generation
4. Text-to-Speech: AI response converted to natural speech via TTS service
5. Audio playback: Trainee hears the AI customer respond in real-time

The entire loop must feel conversational, targeting sub-second latency for the STT→LLM→TTS pipeline. Future roadmap includes video simulation for face-to-face training.

4.8 Voice Session Recording & Storage

► **NEW:** *New section addressing storage of session recordings from voice simulations.*

All voice simulation sessions are recorded and stored for review, evaluation, and compliance purposes:

- Full audio recordings of both the trainee and AI sides of the conversation are captured and stored
- Recordings are encrypted at rest (AES-256) and access-controlled via RBAC
- Configurable retention policies allow companies to set how long recordings are kept (default: 90 days)
- Managers can replay voice sessions alongside the evaluation report for coaching purposes
- Recordings are linked to the corresponding session transcript and evaluation report for unified review

- Storage infrastructure is designed for scalable blob storage (e.g., AWS S3 / GCS) with lifecycle policies for automatic archival and deletion
- Trainees can access their own recordings for self-review; access by others requires appropriate role permissions
- GDPR-compliant: recordings can be deleted on request as part of right-to-erasure workflows

5. Evaluation Engine

The evaluation engine is what separates Usus from simple chatbot role-play. Every interaction is assessed across multiple dimensions with structured, actionable feedback. The goal is for Usus's evaluations to align with how experienced human support managers would judge the same interactions.

5.1 Evaluation Dimensions

Dimension	What It Measures	Weight (Default)
Resolution Accuracy	Did the agent solve the problem correctly? Was the information accurate?	25%
Empathy & Tone	Did the agent acknowledge feelings, show understanding, use appropriate language?	20%
Response Speed	Was the response timely? Did the trainee keep pace without rushing?	10%
Compliance Adherence	Did the agent follow company policies, legal requirements, and security protocols?	15%
Clarity of Communication	Was the response clear, structured, and easy to understand?	10%
De-escalation Skill	Did the agent reduce tension in difficult situations? Or did they make it worse?	10%
Upsell / Cross-sell	Did the agent identify and execute appropriate revenue opportunities?	10%

Dimension weights are configurable per company and per scenario. A compliance-heavy industry (banking, healthcare) can increase compliance weight. A sales-focused team can increase upsell weight.

5.2 Scoring Model

Certification Gate (Onboarding)

- Used during onboarding and hiring evaluation
- Configurable pass/fail threshold (default: 75% weighted score)
- Trainee must pass certification before being cleared for live customer interaction
- Failed scenarios can be retried after reviewing feedback and completing remediation modules

Continuous Scoring (Upskilling)

- Used for ongoing staff development
- No pass/fail gate — continuous scoring tracks improvement over time
- Managers see trend lines, strengths, and areas for improvement
- System recommends specific scenarios to address identified weaknesses

5.3 Feedback Structure

After each simulation, the trainee receives a structured debrief:

- 1. Overall Score: Weighted aggregate across all dimensions (e.g., 82/100)
- 2. Dimension Breakdown: Individual score per dimension with color-coded indicators
- 3. What Went Well: Specific moments where the trainee performed effectively, with excerpts
- 4. What Needs Improvement: Specific mistakes or missed opportunities, with explanation of why
- 5. Ideal Response Examples: What an expert agent would have said at key decision points
- 6. Red Flags: Any compliance violations, misinformation, or inappropriate language detected
- 7. Recommended Next Steps: Suggested scenarios or modules to address weaknesses

5.4 Red Flag Detection

The evaluation engine includes automatic detection and flagging of critical issues:

Red Flag Type	Description	Severity
Compliance Violation	Agent shares restricted info, bypasses security, or violates policy	Critical
Misinformation	Agent provides incorrect product details, pricing, or process steps	High
Inappropriate Language	Rude, dismissive, discriminatory, or unprofessional language	Critical
Security Breach Attempt	Agent falls for social engineering or shares PII inappropriately	Critical
Unauthorized Commitment	Agent makes promises the company cannot fulfill	High
Escalation Failure	Agent fails to escalate when required by policy	Medium

Red flags are surfaced prominently in the feedback report and are visible to managers in the admin dashboard. Critical red flags during certification automatically trigger a fail regardless of overall score.

5.5 Feedback Timing (Configurable)

Mode	When Feedback Appears	Best For
Real-Time Hints	Subtle nudges during the simulation (e.g., “Consider acknowledging their frustration”)	Beginners; guided learning
Post-Simulation Debrief	Full evaluation report after the scenario ends	Standard training; certification
Combined	Light real-time hints + full debrief after	Intermediate trainees
No Hints (Exam Mode)	No feedback during simulation; full report after	Certification gates; hiring evaluation

Feedback timing is configurable per scenario by managers. Certification and hiring evaluations should use Exam Mode to simulate real conditions.

5.6 Remediation Modules

► **NEW:** New section describing what remediation modules look like when a trainee fails certification.

When a trainee fails a certification scenario or scores below threshold on specific dimensions, the system assigns targeted remediation modules. These modules are designed to address the specific weaknesses identified in the evaluation:

Module Structure

- Each remediation module targets one or two evaluation dimensions (e.g., Empathy & Tone, Compliance Adherence)
- Modules consist of a combination of: instructional content (best practices, examples of good vs. bad responses), guided practice scenarios (with real-time hints enabled), and assessment scenarios (exam mode, to verify improvement)
- Modules are shorter than full learning path levels — typically 3–5 focused scenarios per module

Types of Remediation Modules

- **Dimension-Specific Drill:** Focused on a single evaluation dimension. E.g., a trainee who scored poorly on de-escalation receives 3–4 scenarios with increasingly hostile customers, starting with guided hints and ending in exam mode.
- **Red Flag Remediation:** Triggered by critical red flags (compliance violations, security breaches). Includes instructional content on the specific policy violated, followed by scenarios designed to test adherence to that policy.
- **Knowledge Gap Fill:** Triggered when misinformation is detected. Presents the correct information from the company knowledge base, then tests the trainee with scenarios requiring accurate information delivery.
- **Composite Remediation:** For trainees who failed across multiple dimensions, a combined module that sequences dimension-specific drills in priority order.

Completion & Re-certification

After completing a remediation module, the trainee must re-attempt the original certification scenario (or an equivalent scenario of the same difficulty). The system tracks remediation completion and re-attempt scores to measure whether the intervention was effective. Managers receive notifications when trainees complete remediation and are ready for re-certification.

6. Training Knowledge Base

The knowledge base is the foundation that enables Usus to both simulate realistic customers and evaluate trainee performance accurately.

► **REVISED:** *The knowledge base now operates at three layers: Universal CS Knowledge, Online Reviews from Reputable Sites, and Company-Specific Customization.*

6.1 Layer 1: Universal CS Knowledge

Usus is pre-trained on a comprehensive body of customer support principles:

- Communication best practices: active listening, mirroring, clarity, structured responses
- Empathy frameworks: acknowledgment, validation, reassurance techniques
- De-escalation techniques: tone management, pacing, reframing, boundary setting
- Problem-solving frameworks: root cause analysis, systematic troubleshooting
- Compliance fundamentals: data privacy, security protocols, authorization boundaries
- Sales integration: identifying upsell/cross-sell opportunities without being pushy

6.2 Layer 2: Online Reviews from Reputable Sites

► **NEW:** *New layer added to the knowledge base architecture.*

Usus ingests and analyzes real customer feedback from reputable public sources to generate realistic customer personas, emotional patterns, edge cases, and common failure points:

- Sources include: app store reviews, G2, Trustpilot, Capterra, complaint forums, and public social media discussions
- Data is processed to extract: common complaint themes, emotional tone patterns, language and phrasing styles, escalation triggers, and recurring problem types
- This layer bridges the gap between theoretical CS best practices and real-world customer behavior
- The platform maintains an automated pipeline for periodically refreshing this data to keep scenarios current
- Extracted patterns feed directly into scenario generation and customer persona creation

6.3 Layer 3: Company-Specific Customization

At the point of integration, each company uploads their own data to customize the training environment:

Input Type	Purpose	Example
Support Documentation	Teach Usus company processes and resolution paths	Knowledge base articles, SOPs, runbooks
FAQs	Enable accurate information provision during simulations	Product FAQs, billing FAQs, return policies
Product Manuals	Ground simulations in real product details	Feature docs, pricing pages, technical specs
Past CS Interactions	Generate realistic scenarios based on actual customer problems	Chat transcripts, call recordings, email threads

Input Type	Purpose	Example
Company Policies	Enforce compliance during evaluation	Refund policy, escalation matrix, SLAs
Brand Voice Guidelines	Ensure AI customer personas match real customer demographics	Tone guides, terminology preferences

Company data is ingested via a RAG (Retrieval-Augmented Generation) pipeline, stored in a vector database, and retrieved contextually during simulations and evaluations. Each company's data is fully isolated in the multi-tenant architecture.

6.4 Knowledge Base Development Phases

Phase 1: Train Usus as a CS Expert — Before simulating customers, Usus first learns to think like an expert support agent. It is trained on industry best practices, communication research, and proven frameworks. The goal is for Usus to embody ideal CS behavior — correct tone, empathy, clarity, policy adherence, and problem-solving.

Phase 2: Online Review Ingestion — Usus ingests real customer feedback from reputable review sites and public discussion forums. This generates realistic personas, emotional patterns, edge cases, and common failure points that ground the simulation in real-world customer behavior.

Phase 3: Company-Specific Integration — Company documentation, policies, and historical interactions are ingested to create context-aware simulations with accurate policy enforcement and realistic brand voice.

7. Learning Path Engine

Training follows structured learning paths that progress from foundational skills to advanced scenarios. Each stage builds on the last, unlocking new scenarios as competencies are demonstrated.

7.1 Progression Structure

Level	Focus	Scenario Types	Unlock Criteria
1. Foundations	Communication basics, empathy, active listening	Friendly customers, clear problems	Complete all foundation modules
2. Core Skills	Policy adherence, troubleshooting, clarity	Confused customers, multi-step problems	Score $\geq 75\%$ on Level 1 scenarios
3. Pressure	De-escalation, time pressure, emotional customers	Frustrated/angry customers, ambiguous issues	Score $\geq 75\%$ on Level 2 scenarios
4. Advanced	Edge cases, compliance traps, social engineering	Manipulative customers, policy gray areas	Score $\geq 80\%$ on Level 3 scenarios
5. Expert	Complex multi-party, upsell under pressure	VIP clients, compound problems, peer handoffs	Score $\geq 85\%$ on Level 4 scenarios

7.2 Scenario Categories

► **REVISED:** Scenario categories are now configurable by Company Admins. The list below represents the default categories provided by the platform.

Scenarios are organized by category to allow both structured progression and targeted practice. Company admins can create, rename, and reorganize categories to match their organizational structure:

Default categories: Billing & Payments, Technical Support, Account Management, Complaints & Escalations, Sales & Retention, Compliance & Security, Onboarding & Education, Custom Scenarios (company-specific scenarios created by managers).

Company admins can add entirely new categories, merge existing ones, or hide categories that are not relevant to their operations. Changes to categories are scoped to the individual tenant and do not affect other companies on the platform.

7.3 Skill-Based Modules

► **REVISED:** Skill-based modules can now be curated by Company Admins or Platform Admins, not just assigned by managers.

Beyond linear progression, Usus offers targeted skill modules. These can be curated and customized by Company Admins or Platform Admins, assigned by managers, or recommended by the evaluation engine:

- **Empathy Mastery:** Focused practice on tone, acknowledgment, and emotional intelligence
- **De-escalation Bootcamp:** Intensive practice with angry and hostile personas
- **Compliance Deep-Dive:** Scenarios designed to test policy knowledge and security awareness
- **Speed & Efficiency:** Time-pressured scenarios emphasizing quick, accurate resolution

- **Upsell Techniques:** Practice identifying and executing natural sales opportunities

Company Admins can create custom skill modules tailored to their organization's needs, using the Scenario Builder to assemble module content. Platform Admins can curate and publish modules that are available to all tenants as part of the universal library.

8. Admin Dashboard & Management

8.1 Role Hierarchy

Role	Permissions	Typical User
Platform Admin	Full system access; manage tenants; global configuration; curate universal skill modules and scenario library	Usus internal team
Company Admin	Company settings, billing, integrations, data uploads; configure scenario categories; curate company-specific skill modules	Head of Operations, IT Admin
Manager	Create scenarios, set rubrics, view team analytics, assign training	CS Team Lead, Training Manager
Trainee	Access assigned scenarios, view own scores and feedback, progress through paths	CS Agent, New Hire, Candidate

8.2 Manager Capabilities

- **Scenario Builder:** Create custom scenarios with defined personas, problems, difficulty levels, and evaluation criteria
- **Rubric Customization:** Adjust evaluation dimension weights per scenario; add company-specific evaluation criteria
- **Training Assignment:** Assign specific scenarios, modules, or learning paths to individuals or teams
- **Performance Dashboard:** View individual and team-level analytics with trend lines, strengths, and gaps
- **Red Flag Alerts:** Real-time notifications when trainees trigger critical compliance or behavior flags
- **Certification Management:** Set pass/fail thresholds; manage certification workflows for onboarding
- **Voice Session Replay:** Replay recorded voice sessions alongside evaluation reports for coaching

8.3 Analytics & Reporting

In-App Dashboard

- Individual trainee profiles with score history, dimension trends, and learning path progress
- Team-level aggregates: average scores, common weaknesses, top performers
- Scenario analytics: completion rates, average scores, most-failed scenarios
- Red flag summary: frequency, type, and trainee-level breakdown

Exportable Reports

- PDF/CSV performance reports for HR reviews and compliance audits
- Certification reports: who passed, who failed, when, and on which scenarios
- Training completion reports: hours logged, scenarios completed, progress status
- Custom report builder: filter by date range, team, individual, dimension, or scenario

9. Hiring & Candidate Evaluation Module

Beyond training existing staff, Usus can be integrated into the hiring process to evaluate candidates on soft skills, communication ability, and job readiness before they're hired.

9.1 How It Works

8. Company configures evaluation: Select scenarios, set pass threshold, define evaluation criteria
9. Candidate receives link: Unique assessment link sent via email or embedded in ATS
10. Candidate completes simulation(s): Exam Mode — no hints, no feedback during simulation
11. System generates evaluation: Full scored report across all dimensions
12. Hiring team reviews: Report visible in admin dashboard; compare candidates side-by-side

9.2 Assessment Features

- Configurable scenario sets (3–5 scenarios per assessment is recommended)
- Anti-cheating measures: unique AI responses per session, no repeat scenarios
- Time limits per scenario to simulate real work pressure
- Candidate comparison dashboard: rank candidates by overall or per-dimension scores
- Integration-ready: webhook/API for ATS platforms (Greenhouse, Lever, Workable)

10. Multi-Tenancy & Data Architecture

10.1 Tenant Isolation

Usus operates as a multi-tenant SaaS platform. Each company (tenant) has fully isolated data:

- Company-specific knowledge bases, scenarios, and rubrics are siloed per tenant
- Trainee data, scores, and evaluation reports are isolated per company
- No cross-tenant data access under any circumstances
- Tenant-specific configurations: branding, evaluation weights, certification thresholds, scenario categories

10.2 Data Security & Privacy

- All data encrypted at rest and in transit (AES-256, TLS 1.3)
- Role-based access control (RBAC) at every level
- Audit logging for all administrative actions
- Voice session recordings stored encrypted with configurable retention policies
- GDPR-compliant data handling; DPA available for enterprise clients
- SOC 2 Type II and ISO 27001 on compliance roadmap
- Company data (including voice recordings) can be deleted on request (right to erasure)

10.3 Scalability Targets

Phase	Concurrent Users	Infrastructure
Phase 1 (Launch)	Hundreds	Single-region cloud deployment; horizontal scaling ready
Phase 2 (Growth)	Thousands	Multi-region deployment; CDN; load balancing
Phase 3 (Scale)	Tens of thousands	Auto-scaling; dedicated compute for voice; edge caching

11. API & Integration Layer

11.1 API Design

- Standard: RESTful API documented with OpenAPI / Swagger specification
- Authentication: OAuth 2.0 + API key authentication for enterprise integrations
- Webhooks: Event-driven notifications for scenario completion, certification, red flags
- Rate Limiting: Tiered rate limits based on plan tier

11.2 Integration Points

Integration	Type	Purpose
ATS Platforms	API / Webhook	Send candidate assessments to Greenhouse, Lever, Workable
CRM Systems	Read-only data import	Import customer interaction data for scenario generation
LMS Platforms	API	Sync training progress with existing learning management systems
SSO Providers	SAML / OIDC	Enterprise single sign-on (Okta, Azure AD, Google Workspace)
Communication Tools	Webhook	Slack/Teams notifications for managers on certifications and red flags
BI / Analytics	Data export API	Feed performance data into Tableau, Looker, Power BI

12. MVP Scope (Phase 1)

Phase 1 focuses on delivering a functional customer support training simulator with the minimum feature set required to validate the product with real users.

► **REVISED:** Updated to reflect Email/In-Person modes moved to Phase 2, AI customer-only role in MVP, voice recording storage added, and scenario category configuration added.

12.1 In Scope (MVP)

- Text-based simulation (live chat format)
- Voice-based simulation (phone call format) with session recording and storage
- AI plays the customer role only (single AI role per scenario)
- Hybrid scenario generation (scripted framework + AI improvisation)
- Structured process for scenario generation, ranking, and categorization
- 7-dimension evaluation engine with structured feedback
- Post-simulation debrief (full report after scenario ends)
- Remediation modules for trainees who fail certification or score below threshold
- Basic learning path: 3 levels (Foundations, Core Skills, Pressure)
- Manager dashboard: view team scores, assign scenarios, replay voice sessions
- Company-specific data upload (docs, FAQs, policies)
- Three-layer knowledge base: Universal CS Knowledge, Online Reviews, Company-Specific
- Configurable scenario categories by Company Admins
- Multi-tenant architecture with role-based access
- Web application (responsive for mobile browsers)
- Red flag detection for compliance violations and inappropriate language

12.2 Out of Scope (Future Phases)

- Email simulation mode (Phase 2)
- In-Person simulation mode (Phase 2)
- AI as superior/peer roles (Phase 2)
- Real-time hints during simulation (Phase 2)
- Advanced learning path levels 4–5 (Phase 2)
- Hiring/candidate evaluation module (Phase 2)
- ATS and LMS integrations (Phase 2)
- Native mobile apps (Phase 2)
- Custom report builder (Phase 2)
- Video simulation (Phase 3)
- Multi-language support (Phase 3)
- Industry-specific modules beyond CS (Phase 3)

13. Success Criteria

13.1 Product Success Metrics

Metric	Target	How Measured
Simulation Realism	85%+ of users rate simulations as “realistic”	Post-session survey
Evaluation Accuracy	90%+ alignment with expert human evaluators	Blind comparison study
Feedback Usefulness	80%+ of users report feedback improved their skills	Post-training survey
Certification Correlation	Certified agents perform 20%+ better on live metrics	Pre/post certification performance data
System Latency	Text <2s, Voice near-real-time, API <60ms	Application monitoring

13.2 Usus Self-Validation

Before launching, Usus must demonstrate that it can:

13. Consistently identify good vs. bad customer support responses
14. Explain why a response was good or bad with specific, actionable reasoning
15. Role-play customers realistically across all difficulty levels
16. Generate accurate evaluations that match experienced human manager assessments
17. Detect compliance violations and misinformation reliably

14. Team & Ownership

Area	Owner	Responsibilities
Research & Validation	Gideon	Timeline feedback, validating assumptions, feasibility checks
CS Knowledge & Principles	Mabel	Outlining everything a CS should know; knowledge base content
Scenario Design	Mabel	Contextual and real-world CS scenarios; edge case design
Knowledge Gathering Automation	TBD	Automating research and data collection pipelines
User Flow & Design	James	User flow map, brand guidelines, UI/UX, social media elements
Product Direction & Integration	TBD	Platform decisions, scope, priorities, architecture

15. Product Roadmap

► **REVISED:** Updated to reflect items moved between phases.

Phase	Focus	Key Deliverables
Phase 0: Research	Validate assumptions and build knowledge foundation	CS knowledge map, evaluation standards, customer problem sources, initial learning path design, online review ingestion pipeline
Phase 1: MVP	Text + voice simulation with evaluation engine	Simulation engine (text + voice), voice session recording & storage, 7-dimension evaluation, remediation modules, three-layer knowledge base, scenario generation/ranking/categorization process, configurable scenario categories, basic learning path (3 levels), admin dashboard, multi-tenant architecture
Phase 2: Expand	Additional modes, AI roles, hiring module, integrations	Email + in-person simulation modes, AI as superior/peer, candidate evaluation, ATS/LMS integrations, real-time hints, native mobile apps, advanced learning path levels 4–5, custom report builder
Phase 3: Scale	Industry expansion and advanced features	Sales/HR/retail modules, multi-language, video simulation, advanced analytics, marketplace for scenarios

16. Revision Summary (v1.0 → v1.1)

This section summarizes all changes made in this revision:

6. Voice Session Recording & Storage: Added section 4.8 covering storage of voice simulation recordings, encryption, retention policies, and GDPR compliance
7. Simulation Modes: Email and In-Person modes moved from MVP to Phase 2. MVP includes Text Chat and Voice Call only
8. AI Roles: Clarified that MVP will feature Customer role only; Superior/Manager and Peer/Colleague deferred to Phase 2
9. Scenario Generation Process: Added new section 4.5 defining a structured process for generating, ranking, and categorizing scenarios beyond the initial persona set
10. Scenario Categories: Now configurable by Company Admins (section 7.2)
11. Remediation Modules: Added new section 5.6 describing the structure, types, and completion workflow for remediation modules
12. Training Knowledge Base: Restructured from two layers to three layers — Universal CS Knowledge, Online Reviews from Reputable Sites, and Company-Specific Customization (section 6)
13. Skill-Based Modules: Can now be curated by Company Admins or Platform Admins (section 7.3)
14. MVP Scope & Roadmap: Updated sections 12 and 15 to reflect all changes above, ensuring consistency across the document

17. Appendix

17.1 Glossary

Term	Definition
Simulation	An interactive scenario where the trainee interacts with an AI-powered character
Scenario	A defined training situation with context, persona, problem, and evaluation criteria
Debrief	The post-simulation feedback session showing scores and improvement recommendations
Persona	The AI-generated character profile (personality, emotion, difficulty) for a simulated customer
Rubric	The evaluation framework defining how performance is scored across dimensions
Red Flag	An automatically detected critical issue (compliance violation, misinformation, etc.)
Certification Gate	A pass/fail threshold that must be met before a trainee can go live
Remediation Module	A targeted training module assigned after certification failure to address specific weaknesses
RAG	Retrieval-Augmented Generation — technique for grounding AI responses in company-specific data
Tenant	A single company/organization within the multi-tenant SaaS platform
STT / TTS	Speech-to-Text / Text-to-Speech — technologies enabling voice simulation
Session Recording	Encrypted audio capture of voice simulations stored for review and compliance

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